Formulas & Symbol Glossary

Engineering Chemistry (UE26CY151A) · Semester 1 · Unit 1 · Compiled from PESU AV-summary transcripts, class slides, the Unit-1 slide PDF and past ESA papers · 13 Oct 2026

★★★ high · ★★ medium · ★ low/conceptual weight | Not on current slides | Transcript only | Added for clarity | source tags: [S004·8] slide, [T005] transcript, [PDF] 137-page deck, [ESA] past paper

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Symbol glossary

Constants & conversions

Constants: h = 6.626×10^{-34} J s · c = 3×10^{8} m s^{-1} = 3×10^{10} cm s^{-1} · N_A = 6.022×10^{23} mol^{-1} · 1 amu = 1.66×10^{-27} kg · π = 3.14 (papers). **Conversions:** E(J) ÷ hc → cm^{-1}; $\nu = c\cdot\bar{\nu}$; 1 eV = 1.602×10^{-19} J = 8065.5 cm^{-1}; 1 kJ mol^{-1} = 83.59 cm^{-1}.

M1 · Molecular Spectroscopy

T1 · EM radiation & interaction with matter

F1 · Photon energy ★★★

$$E = h\nu$$

Gives Energy of radiation · **Applies** any EM radiation · **Units** J · [S001·5]

F2 · Frequency–wavelength ★★★

$$\nu = c/\lambda$$

Gives Frequency · **Applies** vacuum/air · **Units** Hz · [S001·5]

F3 · Wavenumber ★★★

$$\bar{\nu} = 1/\lambda \; ; \; E = hc\bar{\nu}$$

Gives Wavenumber; energy · **Applies** spectroscopic unit · **Units** cm^{-1} · [S001·5]

F4 · Transition energy ★★★

$$\Delta E = E_2 - E_1 = h\nu$$

Gives Absorbed/emitted quantum · **Applies** two quantised levels · **Units** J (÷hc → cm^{-1}) · [S001]

T2 · Electromagnetic spectrum & regions

No formula for this topic.

T3 · Born–Oppenheimer approximation

F5 · Total molecular energy ★★

$$E = E_trans + E_rot + E_vib + E_elec \approx E_rot + E_vib + E_elec$$

Gives Born–Oppenheimer · **Applies** E_trans negligible · **Units** J · [S003·4]

F6 · Ordering ★★

$$E_rot \ll E_vib \ll E_elec$$

Gives Microwave < IR < UV-Vis · **Applies** — · **Units** — · [S003·4]

F7 · Change in energy ★★

$$\Delta E = \Delta E_elec + \Delta E_vib + \Delta E_rot$$

Gives Electronic transition fine structure · **Applies** — · **Units** J or cm^{-1} · [S007·4]

T3b · Beer–Lambert law

F8 · Beer–Lambert Not on current slides Added for clarity ★

$$A = \log(I_0/I) = \varepsilon cl$$

Gives Absorbance · **Applies** dilute solutions · **Units** dimensionless · [—]

T4 · Rotational (microwave) spectroscopy

F9 · Centre of mass ★★★

$$m_1 r_1 = m_2 r_2 \; ; \; r_0 = r_1 + r_2$$

Gives Positions of atoms about C · **Applies** rigid diatomic · **Units** m · [S004; PDF]

F10 · Distances ★★★

$$r_1 = m_2 r_0/(m_1+m_2) \; ; \; r_2 = m_1 r_0/(m_1+m_2)$$

Gives — · **Applies** — · **Units** m · [PDF]

F11 · Moment of inertia ★★★

$$I = m_1 r_1^2 + m_2 r_2^2 = \mu r_0^2 \text{ with } \mu = m_1 m_2/(m_1+m_2)$$

Gives Moment of inertia · **Applies** rigid rotor · **Units** kg m² · [S004·7; PDF]

F12 · Rotational energy ★★

$$E_r = \tfrac{1}{2} I \omega^2 = L^2/2I$$

Gives Classical rotor · **Applies** — · **Units** J · [PDF]

F13 · Rotational levels (J) ★★★

$$E_J = h^2 J(J+1)/(8\pi^2 I)$$

Gives Quantised levels · **Applies** J = 0,1,2… · **Units** J · [S004·8]

F14 · Rotational levels (cm⁻¹) ★★★

$$\varepsilon_J = BJ(J+1), \; B = h/(8\pi^2 Ic)$$

Gives Level energy; rotational constant · **Applies** c in cm s⁻¹ · **Units** cm⁻¹ · [S004·8]

F15 · Selection rules ★★★

$$\Delta J = \pm 1 \; ; \text{ permanent dipole needed}$$

Gives Allowed transitions · **Applies** gross rule · **Units** — · [S004·10]

F16 · Line positions ★★★

$$\Delta\varepsilon(J \to J+1) = 2B(J+1) \; ; \text{ spacing} = 2B \; ; \text{ levels } 0,2B,6B,12B,20B,30B,42B$$

Gives Rotational spectrum · **Applies** — · **Units** cm⁻¹ · [S004·9-10]

T5 · Vibrational (IR) – SHO

F17 · Hooke's law ★★

$$F = -kx \; ; \; PE = \tfrac{1}{2} k(r - r_eq)^2$$

Gives Parabolic potential · **Applies** SHO · **Units** N ; J · [S005·6-7]

F18 · Oscillation frequency ★★★

$$\nu_osc = (1/2\pi)\sqrt{(k/\mu)}$$

Gives Frequency · **Applies** SHO · **Units** Hz · [S005·8]

F19 · Wavenumber of vibration ★★★

$$\bar{\nu}_{osc} = (1/2\pi c)\sqrt{(k/\mu)}$$

Gives Line position · **Applies** c in cm s^{-1} · **Units** cm^{-1} · [S005·8]

F20 · SHO levels ★★★

$$E_v = (v+½)h\nu_{osc} \; ; \; \varepsilon_v = (v+½)\bar{\nu}_{osc}$$

Gives Levels · **Applies** v = 0,1,2… · **Units** J ; cm^{-1} · [S005·8-9]

F21 · Zero-point energy ★★★

$$\varepsilon_0 = ½\bar{\nu}_{osc}$$

Gives Lowest level (≠ 0) · **Applies** — · **Units** cm^{-1} · [S005·9]

F22 · Selection rule & spacing ★★★

$$\Delta v = \pm 1 \; ; \; \text{spacing} = \bar{\nu}_{osc} \; ; \; \text{one line at } \bar{\nu}_{osc}$$

Gives — · **Applies** dipole must change · **Units** cm^{-1} · [S005·9]

F23 · Force constant ★★★

$$k = 4\pi^2c^2\bar{\nu}^2\mu$$

Gives Bond stiffness · **Applies** μ in kg · **Units** N m^{-1} · [derived]

T6 · Vibrational – anharmonic

F24 · Morse function ★★

$$PE = D_{eq}[1 - \exp\{a(r_{eq} - r)\}]^2$$

Gives Anharmonic potential · **Applies** — · **Units** J · [S006·3]

F25 · Anharmonic levels ★★

$$\varepsilon_v = (v+½)\bar{\omega}_e - (v+½)^2\bar{\omega}_e x_e$$

Gives Converging levels · **Applies** x_e small, positive · **Units** cm^{-1} · [S006·4]

F26 · Transitions ★★

$$\text{Fundamental } 0\to1\text{: } \bar{\omega}_e(1-2x_e) \; ; \; \text{1st overtone } 0\to2\text{: } 2\bar{\omega}_e(1-3x_e) \; ; \; \text{2nd overtone } 0\to3\text{: } 3\bar{\omega}_e(1-4x_e)$$

Gives — · **Applies** Δv = ±1,±2,±3… · **Units** cm^{-1} · [S006·6-7]

F27 · Hot band & ZPE Added for clarity ★★

$$\text{Hot band } 1\to2\text{: } \bar{\omega}_e(1-4x_e) \text{ (left of fundamental)} \; ; \; \varepsilon_0 = ½\bar{\omega}_e(1-½x_e)$$

Gives — · **Applies** higher T · **Units** cm^{-1} · [S006·9; ZPE added]

T7 · Electronic spectroscopy & Franck–Condon

F28 · Electronic spectrum ★★

$$\varepsilon = \varepsilon_{elec} + \varepsilon_{vib} \; ; \; \text{lines labelled } (v',v'')$$

Gives Progression · **Applies** no vibrational selection rule · **Units** cm^{-1} · [S007·6-8]

M2 · Phase Equilibria

T8 · Basic terms & phase rule

F29 · Phase rule ★★★

$$F = C - P + 2$$

Gives Degrees of freedom · **Applies** no external fields · **Units** — · [S008·12]

T9 · Derivation of phase rule

F30 · Variables & constraints ★★★

$$\text{Variables} = P(C-1)+2 \; ; \; \text{constraints} = C(P-1) \; ; \; F = P(C-1)+2 - C(P-1) = C-P+2$$

Gives Derivation · **Applies** — · **Units** — · [S009·7]

F31 · Equilibrium condition ★★★

$$\mu_i^{\alpha} = \mu_i^{\beta} = \mu_i^{\gamma} = \dots$$

Gives Equal chemical potential · **Applies** each component · **Units** J mol^{-1} · [S009·3,7]

F32 · Chemical potential Transcript only ★★

$$\mu_i = (\partial G/\partial n_i)_{\{T,P,n_j\}}$$

Gives Partial molar Gibbs energy · **Applies** — · **Units** J mol^{-1} · [T008 (S009·5 unreadable)]

T10 · One-component system: water

F33 · One-component ★★★

$$F = 3 - P$$

Gives Area F=2, line F=1, point F=0 · **Applies** — · **Units** — · [S009·8]

F34 · Clausius–Clapeyron ★★★

$$dP/dT = \Delta H/(T\Delta V)$$

Gives Slope of curve · **Applies** fusion, vaporisation, sublimation · **Units** Pa K^{-1} · [S009·12-14]

T11 · Two-component: reduced phase rule & Pb–Ag

F35 · Reduced (condensed) rule ★★★

$$F = C - P + 1$$

Gives Constant pressure · **Applies** condensed systems · **Units** — · [S010·2-3]

T12 · Thermal analysis / cooling curves

No formula for this topic.

M3 · Computational Chemistry

T13 · Computational chemistry

No formula for this topic.

Symbol glossary

Clashes: **μ** (reduced mass vs chemical potential), **P** (phases vs pressure), **ε** (energy in cm^{-1} vs molar absorptivity).

Symbol	Meaning in this unit	Unit	Value / note
λ	wavelength	m, nm, Å	
ν	frequency	Hz	
$\bar{\nu}$	wavenumber	cm^{-1}	
c	speed of light	m s^{-1} / cm s^{-1}	3×10^{8} m s^{-1} = 3×10^{10} cm s^{-1}
h	Planck constant	J s	6.626×10^{-34} (slides also use 6.625)
E, ε	energy; ε = energy in cm^{-1} (also molar absorptivity in Beer–Lambert — 3rd meaning!)	J; cm^{-1}; L mol^{-1} cm^{-1}	
ΔE	energy change	J	
J	rotational quantum number	—	0,1,2…
B	rotational constant = $h/8\pi^2Ic$	cm^{-1}	
I	moment of inertia	kg m^2	
μ	reduced mass $m_1m_2/(m_1+m_2)$ — ALSO chemical potential in T8–T9	kg; J mol^{-1}	1 amu = 1.66×10^{-27} kg
m_1, m_2	atomic masses	kg	
r_o	bond length (rigid rotor)	m, Å	1 Å = 10^{-10} m
r_1, r_2	distances of atoms from centre of mass	m	
ω	angular velocity	rad s^{-1}	
L	angular momentum	kg m^2 s^{-1}	
k	force constant	N m^{-1}	
x	displacement	m	
v	vibrational quantum number	—	0,1,2…
ν_osc, $\bar{\nu}$_osc	oscillation frequency / wavenumber	Hz ; cm^{-1}	
$\bar{\omega}_e$	equilibrium vibrational wavenumber (anharmonic)	cm^{-1}	
x_e	anharmonicity constant	—	small, positive
D_eq	depth of Morse well	J	
a	Morse constant	m^{-1}	
r_eq	equilibrium bond length	m	
r_e′, r_e″	bond length in excited / ground state	Å	
v′, v″	vibrational number in excited / ground state	—	
ε′, ε″	energy in excited / ground state	cm^{-1}	
F	degrees of freedom (variance)	—	
C	number of components	—	
P	number of phases — ALSO pressure (clash!)	— ; atm, mmHg	

Symbol	Meaning in this unit	Unit	Value / note
T	temperature	K, °C	
G	Gibbs energy	J	
n_i	moles of component i	mol	
ΔH	enthalpy of transition	J mol^{-1}	
ΔV	volume change of transition	m^3 mol^{-1}	
A	absorbance	—	
I_o, I	incident / transmitted intensity	—	
c, l	concentration, path length (Beer–Lambert)	mol L^{-1}, cm	
N_A	Avogadro constant	mol^{-1}	6.022×10^{23}